

Quantitative Image-Analysis of Blastomeres in the early embryonic development of *Macrostomum Lignano* (Plathelminthes, Macrostomorpha)

- Thesis Summary -

Abstract

The goal of this thesis was to apply instance segmentation (supervised) in a developmental biology context, thereby gain new insights into the spatial distribution of cell nuclei during the early embryonic development of *Macrostomum lignano*, an animal undergoing spiral cleavage. Two *M. lignano* embryos were recorded using light-sheet fluorescence microscopy (SPIM), followed by determining cell lineage trees for both embryos. The model was trained using the dataset from Embryo I and then applied to both embryos. Comparable segmentations of the cell nuclei were obtained for both embryos.

Introduction

Spiral Cleavage

- A conserved, early embryonic development pattern in spiralian animals (Mollusca, Annelida, Platyhelminthes and other taxa)
- Fig. 1 shows an illustration of the spiralian cleavage

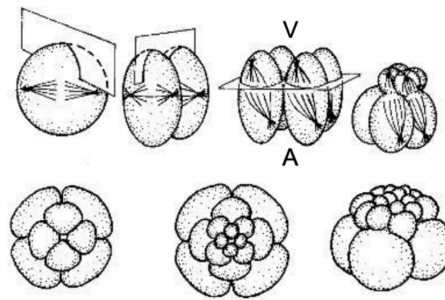


Fig. 1: Illustration of Spiralian Cleavage. Source: [2].

Macrostomum lignano

- a free-living, hermaphroditic flatworm [8]
- model organism for regeneration research
- Fig. 2 shows such flatworm



Fig. 2: *Macrostomum lignano*. Source: <https://earthlingnature.wordpress.com/2020/04/03/friday-fellow-lignanos-macrostomum/> (acc. 12 Sep 2026)

EmbedSeg (Segmentation Model)

- EmbedSeg [4, 5, 3] is an instance segmentation method for microscopy images
- Official GitHub page: <https://github.com/juglab/EmbedSeg>

Mastodon (Cell Lineage Tracking)

- Mastodon [6, 7] is a large-scale tracking and track-editing framework for large, multi-view images
- Official GitHub page: <https://github.com/mastodon-sc/mastodon>

Nomenclature

- I mainly used the standard nomenclature [1] with slight adaptations

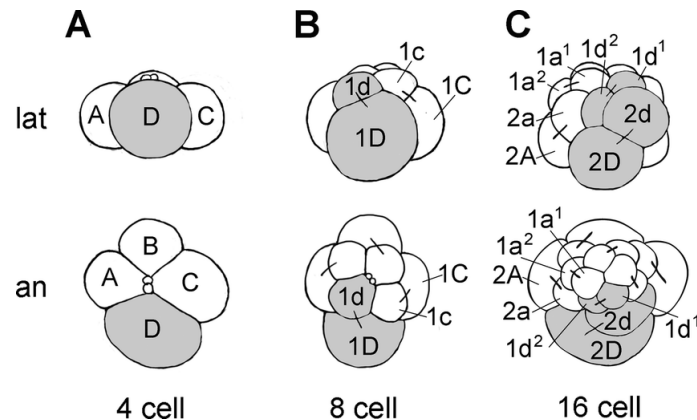


Fig. 3: Illustration of the standard nomenclature of spiralian. Source: <http://placozoa.co.uk/student-2017/Jakob-Lecture-8-Annelida.pdf> (acc. 12 Sep 2026)

Results

History of the Model Training

- History of the training from the model on embryo I
- The generalization was not optimal, still applicable for the purpose of these thesis

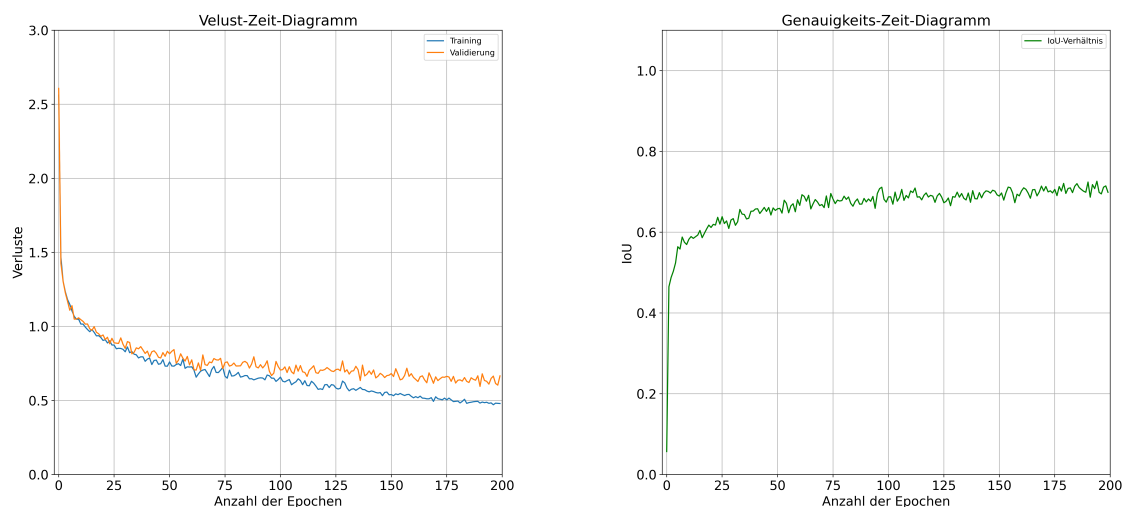


Fig. 4: Scatter plots of training history of the Training of Embryo I. Left: Verlust-Zeit-Diagramm (engl. Loss-Time-Diagramm). Right: Genauigkeits-Zeit-Diagramm (engl. Precision-Time-Diagramm)

Cell Lineage with maximum spatial distribution

- The maximum spatial distribution (in μm^3) of the tracked blastomeres in embryo 2 mapped onto the cell lineage (from tracking).

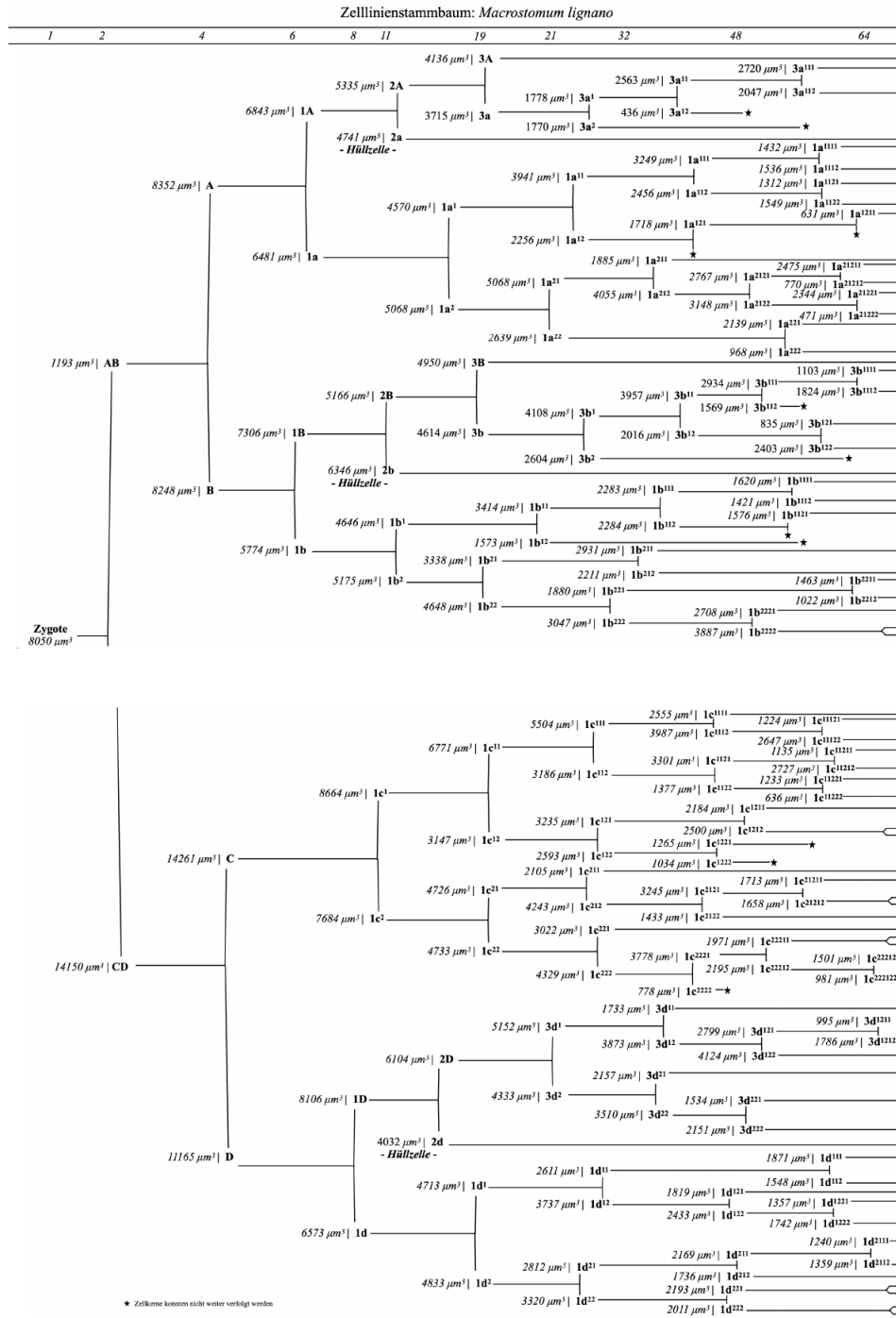


Fig. 5: Scatter plots of training history of the Training of Embryo I. Left: Verlust-Zeit-Diagramm (engl. Loss-Time-Diagram). Right: Genauigkeits-Zeit-Diagramm (engl. Precision-Time-Diagram)

Cell nuclei volume over time from embryo 2

- Fig. 6 illustrates the quantitative image analysis of the blastomeres volume's, just before the third cleavage of second embryo

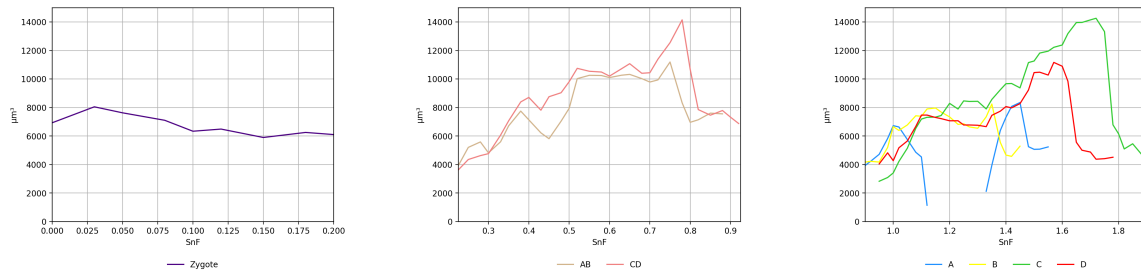


Fig. 6: Left: Zygote before the first splits. Middle: Both blastomeres of the first division (AB & CD). Right: Four blastomeres of the second division (A & B & C & D)

Cell nuclei volume comparison of both embryos over time

- Fig. 7 demonstrates the third split during the early embryogenesis in both embryos ($A > 1A \& 1a$; $B > 1B \& 1b$; $C > 1C \& 1c$; $D > 1D \& 1d$) in of both embryos (Left: Embryo 1 & Right: Embryo 2)

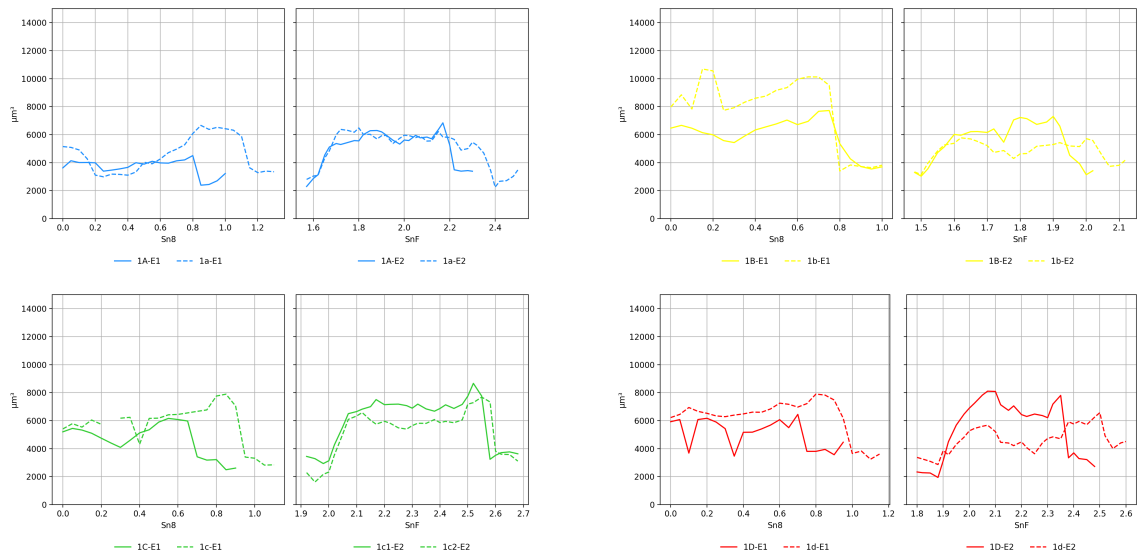


Fig. 7: Embryo Comparison

References

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